

## Parabolas

A quadratic equation has a variable that is squared, or raised to a power of 2. No variable in a quadratic equation can have an exponent greater than 2.

The graph of a quadratic equation forms a **parabola**, a symmetrical curved line. A parabola can open upward or downward.



When you throw a ball into the air, the path of the ball is a parabola.

To graph a quadratic equation, rewrite the equation so that one side equals 0. Then replace 0 with  $y$ .

Solve  $x^2 = 4$ . You already know the equation has two answers: 2 and  $-2$ .

Subtract 4 from both sides.  $x^2 - 4 = 0$

Replace 0 with  $y$ .  $x^2 - 4 = y$ , which can be written  $y = x^2 - 4$

Now you are ready to graph the equation.

### Example

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Graph:  $y = x^2 - 4$

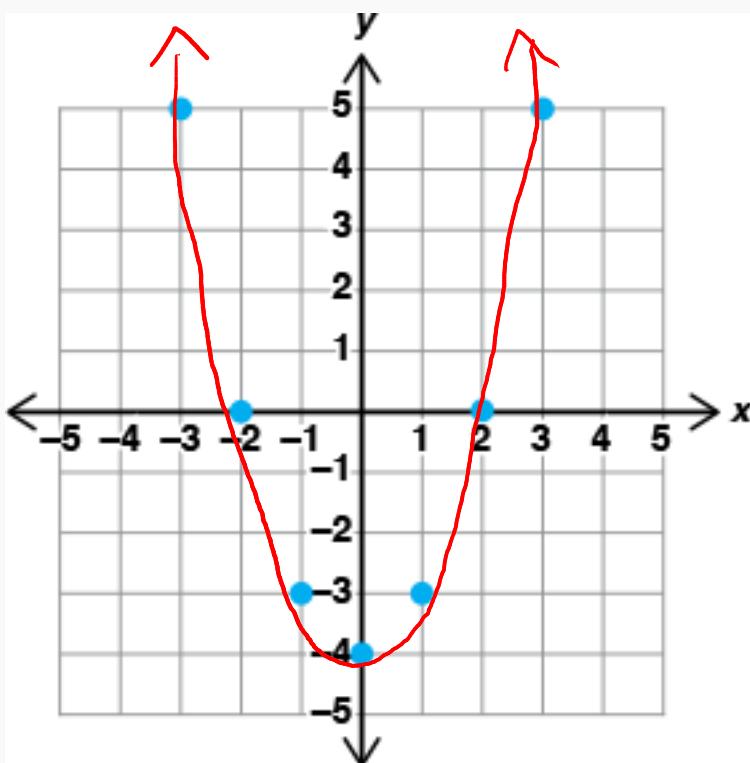
#### Step 1

Make a table of values. Choose values for  $x$ . Substitute them into the equation and solve for  $y$ .

| $x$  | $y$  |
|------|------|
| $-3$ | 5    |
| $-2$ | 0    |
| $-1$ | $-3$ |
| 0    | $-4$ |
| 1    | $-3$ |
| 2    | 0    |
| 3    | 5    |

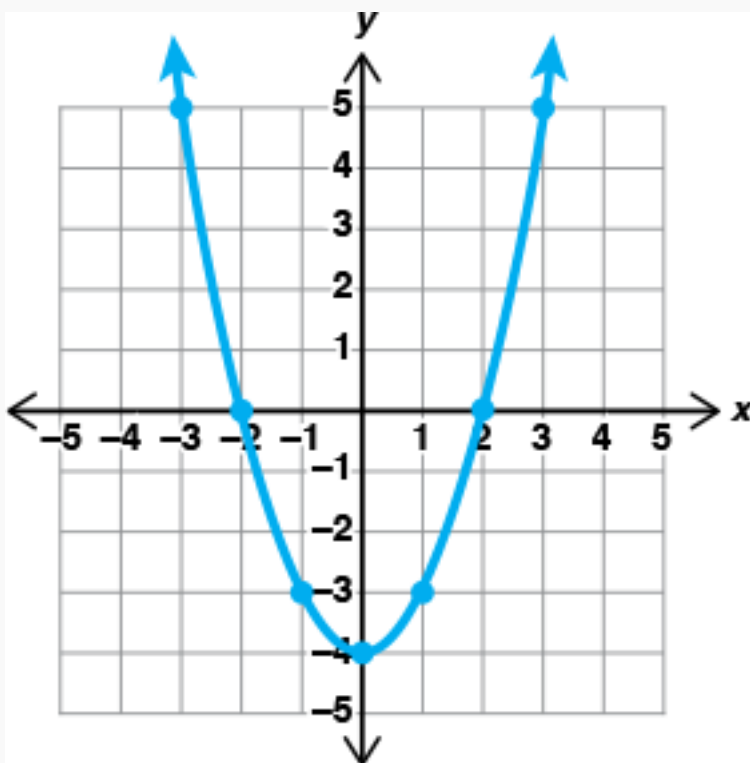
#### Step 2

Plot the points from the table.



#### Step 3

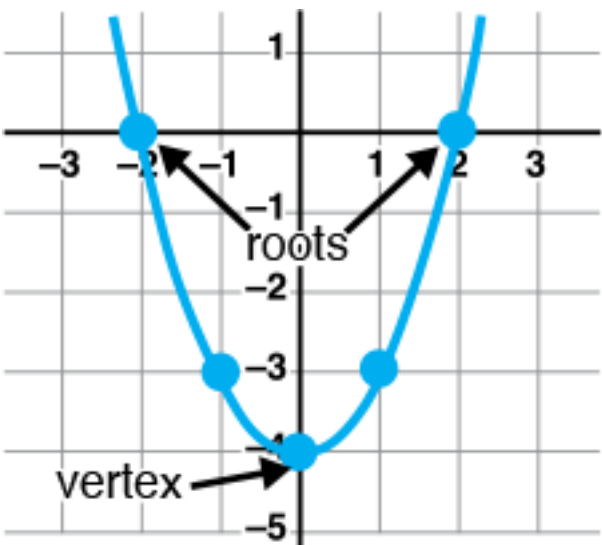
Sketch the parabola.



The graph tells important information about the equation.

The lowest or highest point on a parabola is its **vertex**. (maximum or minimum)

Here, the vertex shows the lower limit of the range. The range of the equation is all real numbers greater than or equal to  $-4$ .



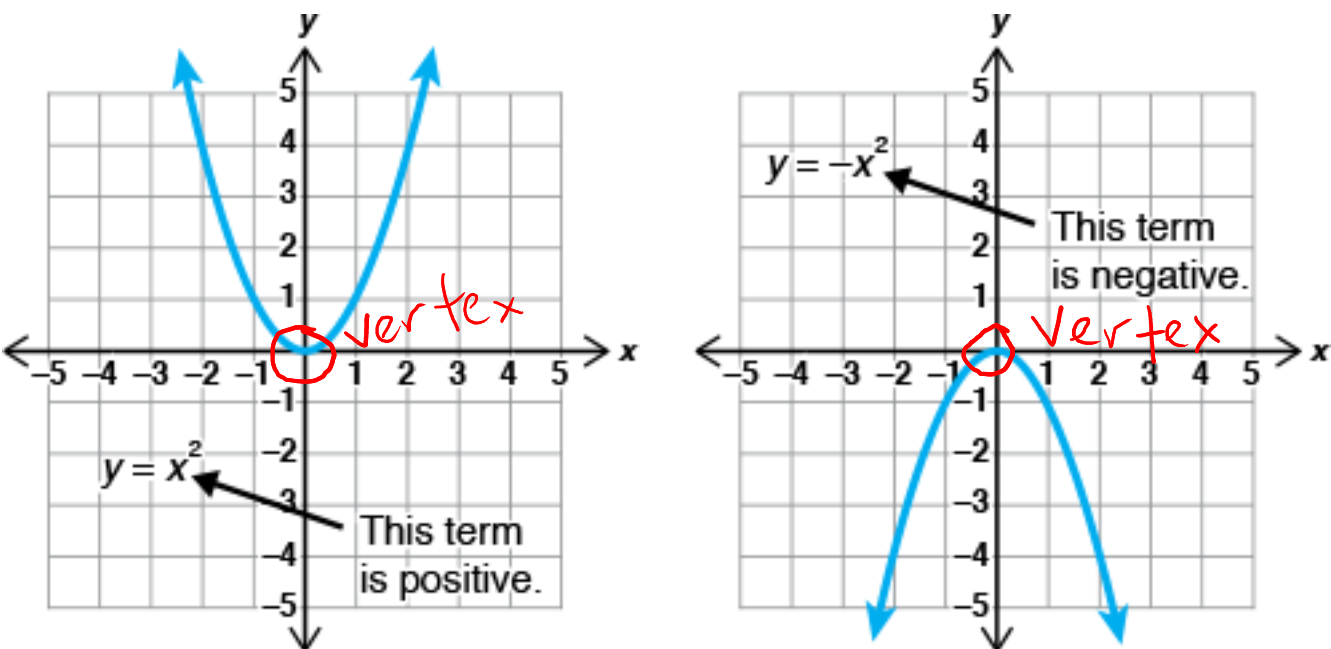
roots are x-intercepts, also called "zeros".

The **roots** are found where the graph crosses the  $x$ -axis. The roots are the solutions to the equation.

The solutions are  $-2$  and  $2$ .

You can also tell which way a parabola will open by looking at the term containing the squared variable.

- If the term is positive, the parabola opens upward.
- If the term is negative, the parabola opens downward.



### Key Terms

completing the square

vertex

parabola

zero property of multiplication

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